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Original Research

DOME: Directional medical embedding vectors from Electronic Health Records

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ABSTRACT

Motivation: The increasing availability of Electronic Health Record (EHR) systems has created enormous potential for translational research. Recent developments in representation learning techniques have led to effective large-scale representations of EHR concepts along with knowledge graphs that empower downstream EHR studies. However, most existing methods require training with patient-level data, limiting their abilities to expand the training with multi-institutional EHR data. On the other hand, scalable approaches that only require summary-level data do not incorporate temporal dependencies between concepts.

Methods: We introduce a DirectiOnal Medical Embedding (DOME) algorithm to encode temporally directional relationships between medical concepts, using summary-level EHR data. Specifically, DOME first aggregates patient-level EHR data into an asymmetric co-occurrence matrix. Then it computes two Positive Pointwise Mutual Information (PPMI) matrices to correspondingly encode the pairwise prior and posterior dependencies between medical concepts. Following that, a joint matrix factorization is performed on the two PPMI matrices, which results in three vectors for each concept: a semantic embedding and two directional context embeddings. They collectively provide a comprehensive depiction of the temporal relationship between EHR concepts.

Results: We highlight the advantages and translational potential of DOME through three sets of validation studies. First, DOME consistently improves existing direction-agnostic embedding vectors for disease risk prediction in several diseases, for example achieving a relative gain of 5.5% in the area under the receiver operating characteristic (AUROC) for lung cancer. Second, DOME excels in directional drug-disease relationship inference by successfully differentiating between drug side effects and indications, correspondingly achieving relative AUROC gain over the state-of-the-art methods by 10.8% and 6.6%. Finally, DOME effectively constructs directional knowledge graphs, which distinguish disease risk factors from comorbidities, thereby revealing disease progression trajectories. The source codes are provided at https://github.com/celehs/Directional-EHR-embedding.

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1. Introduction

Electronic Health Records (EHRs) have attracted significant attention in biomedical research due to their extensive, detailed, and longitudinal documentation of patient clinical information, facilitating a wide range of translational research [1–3]. EHR data can be largely categorized into structured codified data and narrative data. Codified EHR data typically belong to four categories: diagnostic billing codes, laboratory tests, procedure codes, and medication prescriptions. Narrative data consist of a broad range of clinical concepts in clinical notes that can be extracted via natural language processing (NLP) tools and mapped to Concept Unique Identifiers (CUIs) in the Unified Medical Language System (UMLS) [4,5]. These concepts can serve as a complementary source to codified data as they can capture information such as disease symptoms and drug side effects that are often not well coded [6]. Longitudinal observations of large-scale codified and NLP features from diverse patient populations in the EHR hold much promise for generating valuable real-world evidence [7,8], providing insight for precision medicine [9,10], and developing clinical decision support tools [11,12].

Efficiently analyzing EHR data, involving both codified and narrative EHR concepts, however, is challenging due to the high dimensionality of the feature space they span. An effective approach to reducing the feature dimensionality is generating low-dimensional representations of these concepts [13,14]. In recent years, a wide range of concept embedding methods have been proposed, including self-supervised training, knowledge graphs (KGs), co-occurrence mining, etc. These embedding vectors effectively provide prior knowledge on the relations among clinical concepts [15–21] and have been used successfully to improve various downstream tasks, including information retrieval, phenotyping, and disease risk predictions [8,17,19,22–24]. The self-supervised approach devises various auxiliary tasks, typically by predicting the masked parts in sequences, to guide deep language models to learn concepts' representation that captures their clinical semantic relationships [13,25–29]. However, training these models demands extensive sentence-level or patient-level data, such as longitudinal EHRs [16,17,30] or medical notes [18] and it is exceedingly difficult to completely remove privacy-sensitive information from them. This poses risks of privacy leakages. Consequently, due to these privacy issues, it becomes practically unfeasible to engage in collaborative model training across different clinical institutions. Additionally, the embedding vectors generated are typically tailored for predicting clinical outcomes like disease onset or recurrence, rather than for broadly learning general clinical relations among concepts, which limits their use in other related clinical research areas.

KG-based approaches learn concept representations by leveraging established relationships between EHR concepts, e.g. synonymy and may-cause, which are curated by experts in knowledge repositories like the UMLS knowledge base [31–33]. Since they rely on these human-curate relations, these methods face limitations in generalizing the representations to novel diseases or drugs not yet included in existing databases. For instance, when COVID-19 emerged, researchers needed to rebuild domain-specific models to enable predictions involving COVID-19 [34]. Alternatively, a simple yet effective approach is based on low-rank approximations to co-occurrence matrices, which encapsulates the pairwise occurrences of EHR concepts [23,35,36]. Relying solely on aggregated data, these algorithms bypass data-sharing challenges and offer benefits in terms of scalability and privacy preservation.

Despite the success of these embedding techniques, EHR embeddings learned by existing methods do not capture the temporally directional dependencies among healthcare concepts. By "directional dependency", we refer to the relationship in which one concept tends to precede or lead to another concept in time. Such temporal dependency widely exists among the EHR concepts. Specifically, modeling the directional disease-disease relationships would distinguish risk factors from comorbidities, uncovering disease progression trajectories [37,38]. For instance, although prevalent embedding vectors proficiently elucidate the pronounced correlation between hypertension and coronary artery disease (CAD) diagnoses, attributed to their frequent co-occurrence in EHRs, they fail to capture their directional dependency, in other words, whether hypertension tends to precede CAD diagnosis or the converse. Such directional interdependence is similarly observed in drug-disease interactions, which discern whether a pharmaceutical agent ameliorates a condition or induces it as an adverse effect, a distinction crucial for pharmacovigilance efforts. For instance, "paroxetine" and "disruptive behavior disorder" exhibit frequent co-occurrence within EHR datasets, leading to a notable semantic similarity between them. Nevertheless, existing embeddings largely fail to shed light on whether "paroxetine" serves as a treatment for "disruptive behavior disorder" or, contrarily, contributes to its onset [39,40]. Modeling such temporally directional dependencies is indispensable across a spectrum of healthcare research domains, including the identification of disease subtypes [41], risk prediction [42,43], disease progression modeling [44], etc.

To capture such temporal dependencies, existing embedding vectors generally require additional learning models, such as LSTM, and large-scale annotated data for supervision [8,17,18,23,24]. In addition to the limited interpretability, these methods can hardly handle novel healthcare concepts that lack such annotations, for example, the mortality of COVID-19 patients [45]. To the best of our knowledge, there are no relevant studies in the clinical domain literature. The only related work is an unsupervised word embedding learning method called directional skip-gram [46] which, instead of focusing on the directional properties, aims to improve the word semantic embedding by additionally predicting the left and right directions of the context words based on sentence-level data.

In this paper, we introduce the DirectiOnal Medical Embedding (DOME) algorithm, which is designed to effectively capture the directional relationships among EHR concepts by learning from summary-level EHR data. DOME represents each concept using three distinct embedding vectors: a semantic embedding vector that conveys the concept's meaning, and two context embeddings, prior and posterior, that, respectively, describe the concept's tendency to appear before or after other concepts. To achieve this, DOME consists of three steps. First, an asymmetric co-occurrence matrix is aggregated from patient-level EHR data. In contrast to the typically-used symmetric co-occurrences [35,47], the asymmetric co-occurrence matrix distinguishes the pairwise prior co-occurrences from the posterior ones. Next, two Positive Pointwise Mutual Information (PPMI) matrices are computed from the asymmetric co-occurrence matrix to represent the prior and posterior dependencies, respectively. Lastly, joint matrix factorization is performed on the two PPMI matrices to derive the semantic embeddings, prior context embeddings, and posterior context embeddings.

The capability of DOME in enhancing EHR-based research is showcased through three representative tasks: disease risk prediction, drug-disease relationship inference, and KG extraction. The task of predicting disease risk holds significant clinical value for patient management and serves as a common metric for evaluating the quality of embedding vectors [15,17,19,22]. This process primarily involves analyzing the temporal relationships between recorded medical concepts and the target disease within an individual's EHR. Empirical evaluations across five distinct phenotypes consistently reveal the superior performance of DOME compared to existing methodologies, in both supervised and unsupervised scenarios. Notably, DOME has the unique ability to identify risk factors and assess disease risks in an unsupervised manner, an essential feature for researching emerging diseases with scarce prior knowledge. In drug-disease relationship inference, DOME effectively differentiates between the prior and posterior temporal dependencies of drug indications and side effects, significantly outperforming existing embedding vectors. Furthermore, we derive directional KGs

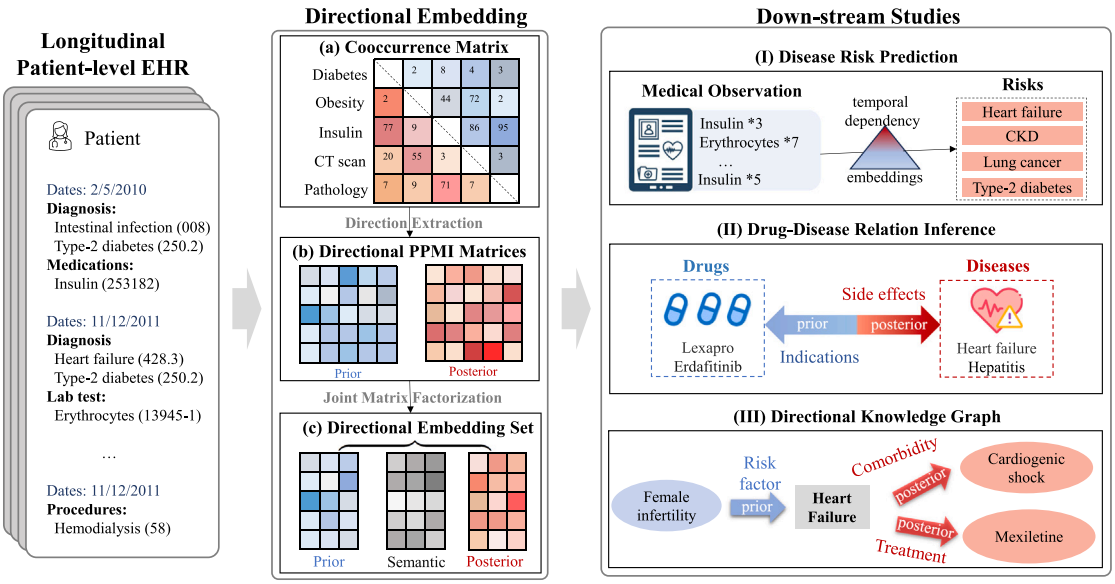

Fig. 1. Overview of the study. In (a), patient-level EHR data are first aggregated into an asymmetric summary-level co-occurrence matrix which summarizes the pairwise correlations between EHR concepts. This co-occurrence matrix is then transformed into two directional PPMI matrices (b). Through joint matrix factorization, each concept is represented by a set of three embedding vectors (c). These embedding vectors are then evaluated for their effectiveness on disease risk prediction (I), drug-disease relationship inference (II), and the extraction of directional knowledge graphs (III).

from DOME without human supervision, uncovering the temporally directional relationship between various EHR concepts. In the case studies on six representative phenotypes, the directional KGs prove to effectively identify disease risk factors, complications, and treatments.

Statement of Significance

Problem: Generating low-dimensional medical embedding vectors from large-scale electronic health records that effectively encode temporal dependencies between medical concepts while preserving patient privacy remains an underexplored challenge.
What is Already Known: Existing embedding techniques like word2vec and GloVe successfully capture the clinical semantics of medical concepts but are unable to reveal their temporal relationships.
What This Paper Adds: This study presents a technique for condensing the summary-level EHR data into low-dimensional, directionally informed embeddings. These embedding vectors facilitate EHR-based translational research, as demonstrated through three applications: predicting disease risk, inferring drug-disease relationships, and extracting knowledge graphs.

2. Method

DOME learns directional embedding vectors from a summary-level co-occurrence matrix of EHR data, which encapsulates pairwise temporally directional relationships between EHR concepts. For each concept, a set of three embedding vectors is generated to correspondingly delineate its semantic, prior context, and posterior context. An overview of this study is provided in Fig. 1. We first describe the process of generating directional embedding vectors in Section 2.1. Then we present how DOME enhances disease risk prediction in Section 2.2.1 and directional relationship inference in Section 2.2.2. Finally, we discuss the data-driven derivation of directional medical KGs in Section 2.3.

2.1. DOME: Learning directional embedding vectors

DOME consists of three major steps: (i) aggregating asymmetric co-occurrence matrix from patient-level EHR data, (ii) computing two PPMI matrices, and (iii) deriving semantic, prior context, and posterior context embedding vectors through joint matrix factorization.

2.1.1. Data source

Patient-level EHR data are obtained from the Mass General Brigham (MGB) Biobank, which is comprised of longitudinal EHR data from 120,032 patients with a median/mean observation window of 12.5/13.1 years. There are five data domains. Four of them are structured codified data (diagnosis, procedures, lab measurements, and medications), which are, correspondingly, mapped to common ontologies, including PheCode (https://phewascatalog.org/phecodes), CCS, LOINC (https://loinc.org/), and RxNorm (https://www.nlm.nih.gov/research/umls/rxnorm/). The fifth domain, the narrative concepts, are mapped to the UMLS CUIs by processing all free-text narrative notes with Narrative Information Linear Extraction (NILE) NLP software [4].

2.1.2. Aggregation of asymmetric co-occurrence matrix

In the first step, we aggregate the patient-level EHR data into a summary asymmetric co-occurrence matrix by scanning through patient visits longitudinally with a T-day sliding window. Specifically, as shown in Fig. 2(a), with the target concept of interest (in this case, "diabetes") centered in the window, the concepts that occur before it within $T/2$ days, like "obesity", are denoted as the prior contexts and those occurring after it within $T/2$ days, like "insulin", are treated as the posterior contexts. By sliding the window through all the visits for each concept and accumulating the counts across all patients, we obtain a $|V|$-by-$|V|$ asymmetric co-occurrence matrix, denoted as N, that summarizes the pairwise directional co-occurrences. Here, $|V|$ represents the size of the vocabulary V. Specifically, the element at the ith row and the jth column, denoted as $N(w_i \rightarrow c_j)$, is the observed total occurrence of the ith concept of the vocabulary preceding the jth concept when the ith concept is treated as the target, while the jth row and the ith column is the total occurrence of the ith concept succeeding the jth concept, denoted as $N(c_j \rightarrow w_i)$.

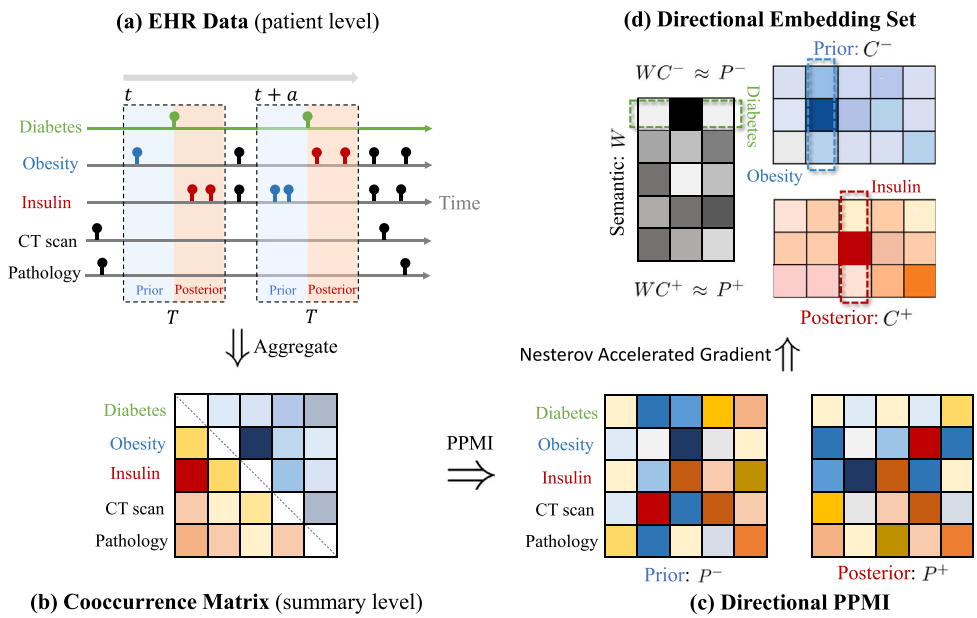

Fig. 2. Procedures of DOME. We first aggregate patient-level longitudinal EHR data using a sliding time window of T into an asymmetric co-occurrence matrix, upon which prior and posterior PPMI matrices are simultaneously computed. Then, joint matrix factorization is performed on the two PPMI matrices to derive a set of three directional embedding vectors, including the shared semantic embeddings and the directional context embeddings.

To preserve the pairwise directional dependency between concepts, we require firstly for each patient, the concepts within the same visit of the target concept to be excluded as the contexts from the co-occurrence matrix, i.e., only the visits prior or posterior it are treated as contexts. Secondly, for each concept pair, only the first co-occurrence observation is counted for each patient. For example, as shown in Fig. 1(a), with "diabetes" as the target, only its first co-occurrence with "obesity' in this patient at window t, namely "obesity" before "diabetes", is included and the observation at the later window $t+a$, "obesity" after "diabetes" is discarded. This is due to the recurrence nature of each concept in longitudinal EHR data, and in this case, "diabetes" would be repetitively observed as priors and posteriors to "obesity", which would weaken directional properties.

2.1.3. Directional PPMI

In the second step, we compute two PPMI matrices that are asymmetric, accounting for the different temporal dependencies (see derivation in Supplementary Note 1 and 2). It has been demonstrated that the matrix factorization of one symmetric PPMI matrix corresponds to the skip-gram with a negative sampling algorithm, which generates EHR embedding vectors that capture their semantic relatedness [23,48]. To further capture the directional properties of the relationship, we instead compute two asymmetric PPMI matrices, P^- and P^+, to correspondingly encode the prior and posterior pairwise dependencies between EHR concepts. The elements of P^- and P^+, are defined as

$$P_{ij}^- = \max\left(\log\left(\frac{N(c_j \to w_i)|D^-|}{N(w_i^-)N(c_j^-)}\right), 0\right) \tag{1}$$

and

$$P_{ij}^+ = \max\left(\log\left(\frac{N(w_i \to c_j)|D^+|}{N(w_i^+)N(c_j^+)}\right), 0\right), \tag{2}$$

where

$$N(w_i^+) = \sum_j N(w_i \to c_j), \quad N(w_i^-) = \sum_j N(c_j \to w_i),$$
$$N(c_j^+) = \sum_i N(w_i \to c_j), \quad N(c_j^-) = \sum_i N(c_j \to w_i),$$

and D^- and D^+ correspondingly denote the collection of observed occurrences of prior-target and posterior-target pairs.

2.1.4. Joint matrix factorization

In the third step, DOME generates a set of three embedding vectors: semantic embedding $W \in \mathbb{R}^{|V|\times p}$, prior context embedding $C^- \in \mathbb{R}^{p\times|V|}$, and posterior context embedding $C^+ \in \mathbb{R}^{p\times|V|}$, where p denotes the dimension of the embedding vectors. Three of them together comprehensively describe the temporal dependency between medical concepts. The semantic embedding W represents the meaning of concepts, similar to most existing embedding methods [35,47]. The context embedding vector C^-(or C^+) denotes the tendency of a concept to occur prior (or posterior) to the target as a context. Specifically, the likelihood of concept i occurring prior to concept j is given by:

$$P(i \to j) = \sigma(W_j C_i^-), \tag{3}$$

where $\sigma(x) = 1/(1+\exp(-x))$, W_j is the semantic embedding vector of concept j and C_i^- is the prior context embedding vector of concept i. Similarly, the likelihood of concept i occurring posterior to concept j is given by:

$$P(j \to i) = \sigma(W_j C_i^+), \tag{4}$$

where C_i^+ is the posterior context embedding vector of i.

To derive the set of directional embedding vectors, we conduct a joint matrix factorization on the prior PPMI matrix, P^-, and posterior one, P^+, to obtain the shared semantic embedding vectors, W, along with the corresponding prior and posterior context embedding vectors C^- and C^+. In comparison to matrix factorizations on P^- and P^+ separately, which lead to two distinct "semantic" embedding W, the shared semantic W and the contextual prior-posterior context C^-/C^+ together enables more flexible modeling of the three relationships simultaneously. The joint matrix factorization is achieved by optimizing W, C^+ and C^- to satisfy: $WC^+ \approx P^+$ and $WC^- \approx P^-$. The training objective is:

$$\min_{W,C^-,C^+} \|WC^+ - P^+\|_F^2 + \|WC^- - P^-\|_F^2. \tag{5}$$

The joint matrix factorization on the two matrices is achieved by optimizing W and C iteratively using Nesterov Accelerated Gradient (NAG) [49], as shown in Algorithm 1 (for a detailed derivation, refer to Supplementary Note 3).

Data: directional PPMI P^+ and P^-
Result: directional embedding vectors W, C^+ and C^-
Initialization
while *not converge* **do**

$$\tilde{W}_{(t+1)} \leftarrow \mu\tilde{W}_{(t)} - \epsilon\Big[(W_{(t)} + \mu\tilde{W}_{(t)})C^+_{(t)}(C^+_{(t)})^T - P^+(C^+_{(t)})^T + (W_{(t)} + \mu\tilde{W}_{(t)})C^-_{(t)}(C^-_{(t)})^T - P^-(C^-_{(t)})^T\Big]$$

$$W_{(t+1)} \leftarrow W_{(t)} + \tilde{W}_{(t+1)}$$

$$\tilde{C}^+_{(t+1)} \leftarrow \mu\tilde{C}^+_{(t)} - \epsilon\Big[-W^T_{(t+1)}P^+ + W^T_{(t+1)}W_{(t+1)}(C^+_{(t)} + \mu\tilde{C}^+_{(t)})\Big]$$

$$C^+_{(t+1)} \leftarrow C^+_t + \tilde{C}^+_{t+1}$$

$$\tilde{C}^-_{(t+1)} \leftarrow \mu\tilde{C}^-_{(t)} - \epsilon\Big[(-W^T_{(t+1)}P^- + W^T_{(t+1)}W_{(t+1)}(C^-_{(t)} + \mu\tilde{C}^-_{(t)})\Big]$$

$$C^-_{(t+1)} \leftarrow C^-_{(t)} + \tilde{C}^-_{(t+1)}$$

end

Algorithm 1: Optimization of DOME. Joint matrix factorization is performed on prior PPMI matrix P^- and posterior P^+ to derive the shared semantic embedding W, the prior embedding C^- and the posterior embedding C^+.

2.2. Validation studies for DOME

The promises of DOME are highlighted in three downstream studies. First, we predict disease risks by leveraging embedding vectors to mine the temporal dependencies between medical concepts and target diseases. Then, we infer drug-disease relationships that distinguish drug indications and side effects. Lastly, we derive medical KGs that uncover directional relationships between various medical concepts.

2.2.1. DOME for disease risk prediction

We evaluate the quality of the directional embedding vectors in both unsupervised and supervised disease risk predictions. Disease risk prediction is a typical downstream task for assessing the quality of embedding vectors [15,17,19,22]. Accurate risk predictions not only help patients take precautions but also facilitate healthcare administration. For instance, healthcare efficiency can be improved by directing hospitals to optimize patients' appointments so that those with higher risks are prioritized. An accurate risk prediction requires modeling the temporal dependency between observed EHR data and the target diseases. Most existing medical embedding methods are limited in the supervised scenario [17,20,22], where extensive annotations are required for model training. Besides supervised learning, we also consider risk prediction in the unsupervised scenario, i.e., training without annotated patient-level data, which is important for novel diseases with limited knowledge.

Unsupervised risk prediction. We derive patients' disease risk in an unsupervised manner by leveraging the directional embedding vectors to capture the temporal dependency between observed EHR concepts and target diseases. Specifically, we first select a set of medical concepts that are relevant to the target disease, d, using the medical knowledge graph KESER [23], denoted as $\mathcal{M}_d$. Then, given an observation window of longitudinal EHR data in the past 12 months, we obtain the risk score of the observed concept i, $\sigma(W_iC^+_d)$ in (4) (recall that W_i is the semantic embedding of the concept i and C^+_d is the posterior context embedding of the disease d). Next, we estimate the risk of the disease d by aggregating the risk scores across all the observed concepts weighted by their counts within the observation window

$$P(Y_d = 1) = \frac{1}{n_{d'} + \sum_{i\in\mathcal{M}_d} n_i/h}\left(\sigma(W_{d'}C^+_d)n_{d'} + \sum_{i\in\mathcal{M}_d}\sigma(W_iC^+_d)n_i/h\right), \quad (6)$$

where Y_d denotes the onset status of the target disease, d, n_i is the observed count of the concept i, d' denotes the most indicative concept whose semantic embedding vector has the largest inner-product with the posterior context embedding vector of the target disease, namely $d' = \text{argmax}_{d'\in V} W_{d'}C^+_d$, and h is the hyper-parameter that prevents the risk score from being overwhelmed by the frequent observation of trivial concepts as in [18].

Supervised risk learning. We also consider the scenario where a few labeled data are available to train a risk prediction model. Specifically, we learn a multi-layer perception (MLP) network as in [17] to aggregate the risk scores of all the observed concepts in a learnable manner. We first compute the weighted risk score for each concept i, $\sigma(W_iC^+_d)n_i$ in the observation window. Then the risk scores are fed as input into the MLP model to predict the risk of disease d. We use "0" or "1" to denote the absence or occurrence of the disease d in the next T months and train the MLP network using a binary cross-entropy loss. The architecture of the MLP network and training details are provided in Supplementary Note 4.

Validation study details. The risk prediction is performed based on the patient-level data from the MGB Biobank cohort which contains EHR data of 120,032 patients. Based on a one-year observation window, we predict a patient's disease onset risk within the next T months [16–18,50]. We treat the corresponding diagnostic codes of the target diseases as the label [16,17] and evaluate the performance on five representative chronic phenotypes: heart failure with reduced EF (HFrEF, Phecode:428.3), lung cancer (LC, Phecode:165.1), chronic kidney disease (CKD, Phecode:585.3), type-2 diabetes (T2D, Phecode:250.2), and Dementias (DEM, Phecode:290.1). We filter the patients without diagnostic code of the target disease observed in the EHR data and obtain 5,287 patients with HFrEF, 5,011 with LC, 14,560 with CKD, 19,648 with T2D, and 5,171 with DEM. To demonstrate the cross-site generalizability of the embedding vectors, we perform onset risk prediction of DEM and evaluate the performance using 3000 patients randomly sampled from the University of Pittsburgh Medical Center (UPMC, details provided in Supplementary Note 6). We simultaneously consider short-term predictions with $T = 6$, namely the risk of the next 6 months, and long-term $T = 12$, namely the next 12 months. To this end, we divide each patient's longitudinal EHR data into calendar-year windows and predict the disease risks based on EHR observations from the previous year. For example, we predict a patient's onset risk of T2D in 2016 based on the EHR data in 2015. In the supervised scenario, the yearly observation windows of all patients are randomly split by 70%, 20%, and 10% for training, validation, and test, while in the unsupervised scenario, all patients are used for evaluation. The performance is evaluated with Area Under the Receiver Operating Characteristic Curve (AUROC) as in [16,17].

To demonstrate the significance of DOME, it is benchmarked against five embedding learning techniques: CODER [31], SapBERT [32], PubMedBERT [28], SGNS [47], and GloVe [35]. CODER, SapBERT, and PubMedBERT learn embedding vectors by finetuing pre-trained deep language models. Both SGNS and GloVe derive embedding vectors from the summary-level EHR data. SGNS performs matrix factorization on the co-occurrence matrix of EHR concepts, while GloVe proposes paying additional attention to local context to capture fine-grained semantic regularities. We derive the SGNS and GloVe embedding vectors from the same MGB EHR data. In CODER, SapBERT, PubMedBERT, SGNS, and GloVe, there is only one semantic embedding vector generated for each concept, based on which we compute the risk score of each concept as DOME using (6). In addition, in the supervised scenario, we further compare to BEHRT [16], a BERT-based [26] model able to embed patients' longitudinal EHR data. We obtain an embedding vector for each patient, upon which we train an MLP network for risk prediction. All the methods compared use the same features for risk prediction.

2.2.2. DOME for directional relationship inference

DOME is further examined in the inference of drug-disease relations. The aim is to differentiate between the indications and side effects of drugs based on directional embeddings. Typically, side effects are observed posterior to the administration of corresponding drugs, while indications are prior to the prescription of the drugs. Drug side effects are severely under-reported [51], making their management complicated and costing up to 30.1 billion dollars annually [52] in the US. The longitudinal EHR data record the directional relationship between drugs and side effects, thereby facilitating side effect discovery to improve pharmacovigilance. We assess DOME's capability to discover side effects in both unsupervised and supervised scenarios.

Unsupervised relationship inference. We infer pairwise drug-disease relationships by computing their directional dependency scores based on the semantic embedding vectors of the drugs and the context embedding vectors of diseases. Specifically, as shown in (4) and (3), we formulate the probability of a drug d having a side effect s as $P(d \rightarrow s) = \sigma(W_d C_s^+)$. Similarly, we compute the possibility of a drug d having an indication i as $P(i \rightarrow d) = \sigma(W_d C_i^-)$.

Supervised relationship learning. In the supervised scenario, we treat the drug-disease relation inference as a binary classification task by training an MLP network to predict the existence of the therapeutic or side effects relationships as in [24]. For a given drug-disease pair from the entire drug-disease set, $\mathcal{R}$, say, $(dr, di) \in \mathcal{R}$, the MLP network receives the drug's semantic embedding vector, W_{dr}, and the disease's embedding set, W_{di}, C_{di}^-, and C_{di}^+. The MLP relationship learning network consists of two parts: embedding fusion (EF) and relationship predictor (RP), both of which consist of feedforward layers. The EF first fuses the drug embedding, W_{dr} and disease embeddings, W_{dr}, C_{dr}^-, and C_{dr}^+ separately to correspondingly obtain three fused representations which are combined via the predictor to infer the relationship. The detailed structures of the EF and RP are provided in Supplementary Note 5. We train the MLP using a cross-entropy loss $\mathcal{L}_{\text{rela}}$:

$$\mathcal{L}_{\text{rela}} = -\frac{1}{|\mathcal{R}|} \sum_{(di,dr)\in\mathcal{R}} y_{(di,dr)} \log(\text{MLP}(W_{dr}, W_{di}, C_{di}^-, C_{di}^+) + (1 - y_{di,dr})) \times \log(1 - \text{MLP}(W_{dr}, W_{di}, C_{di}^+, C_{di}^-)), \quad (7)$$

where $y_{di,dr}$ (0 or 1) is the label of the drug-disease pair and $|\mathcal{R}|$ denotes the total number of annotated drug-disease pairs.

Validation study details. **Drug side effects**: We evaluate the performance based on the drug-side-effect pairs collected in the SIDER 4.1 dataset [53]. It contains 1,430 drugs, mapped to Drugbank ID, and 5,880 side effects represented as CUIs. Following [24], we first map drugs from the Drugbank ID to the RxNorm codes to align with the concept coding in the EHR data. This results in 895 shared drugs between SIDER 4.1 and our MGB EHR embeddings. We focus on side effects of medium frequencies as in [54] by requiring the side effects to be reported by at least 2 and no more than 30 drugs. As a result, we obtain a total of 22,367 pairs of drug side effects, covering 642 side effects and 895 drugs. Each drug is reported with 24.99 side effects on average. Discovering drug side effects is challenging as their semantic relationship could be complex. For example, some side effects such as "diarrhea" and "fatigue" are shared by drugs of different categories such as cancer and psychiatric, while some are drug-specific, for example, "alopecia" typically results from cancer drugs.

Drug indications: We evaluate the performance by leveraging the collected drug-indication annotations from the PrimeKG-indication [24, 55]. We map the drugs to RxNorm codes and indications to CUIs correspondingly. We obtain 3,379 drug-indication pairs, consisting of 242 indication CUIs and 1,171 drugs. On average, there are 2.89 reported indications for each drug. Different from side effects, the indications tend to be more drug-specific and observed with a stronger co-occurrence with the corresponding drugs in the EHR data.

In the unsupervised scenario, all the drugs with reports are used for evaluation while in the supervised scenario, we randomly select 70%, 20%, and 10% drugs for training, validation, and test [56,57]. Similar to the risk prediction, we compare DOME with CODER [31], SapBERT [32], PubMedBERT [28], SGNS [47], and GloVe [35]. For each drug, the reported side effects or indications are treated as positives and the unreported pairs are treated as negatives. We evaluate the results using AUROC and recall@k (the recall at the top k predictions) both of which are averaged from 5 random train, validation, and test splits.

2.3. Directional knowledge graph extraction

We derive directional KGs that unveil temporal dependencies between medical concepts in an unsupervised manner by exploiting their directional dependencies. In the graph, temporal relatedness is represented by directional edges linking a prior to the posterior concept. For the prior relationship, we focus on the discovery of disease risk factors, while for the posterior, we investigate the complications and treatments. To construct the KG, we estimate the tendency of a concept i preceding a given target concept j by the prior score, (3), and the tendency of a concept j leads to i by the posterior score, (4). Such directional knowledge aligns with the principle of "granger-causality" [58] and can facilitate the identification of disease risk factors and complications.

Validation study details. The validity of the resulting knowledge graph is evaluated on six representative phenotypes, including heart failure with reduced EF (HFrEF Phecode:428.3), lung cancer (LC, Phecode:165.1), type 2 diabetes (T2D, Phecode:250.2), Parkinson's disease (PD, PheCode:332), rheumatoid arthritis (RA, PheCode:714.1), and chronic kidney disease (CKD, PheCode:585.3). We provide both quantitative evaluations by curating a dataset containing pairwise directional relationships and qualitative evaluations through comprehensive literature review.

Quantitative evaluation of the derived KGs requires comparing them to "gold-standard" directional relations. Since there are no publicly available datasets containing such directional relations, we thus curate a dataset consisting of pairwise directional relationships by combining semantic embeddings and literature review. Specifically, for a given target concept, we first nominate a set of candidate concepts by taking the union of the most relevant concepts based on the corresponding embedding vectors of DOME, SGNS, GloVE, and CODER. In DOME, we generate the prior candidates by selecting the top 100 concepts having the highest prior similarity in (3) to the target while for SGNS, GloVE, and CODER, we use the top 100 concepts with the highest cosine similarity to the target. We then filter these nominations using the GPT-4 [59] and manually remove the relation pairs if we do not find any supporting literature. Finally, we further refine the directional relation by adding pairs extracted from the literature or Mayo Clinic [60] (see Supplementary Note 7 for details). For posterior relation curations, the pipeline remains the same except that the prior score in DOME is replaced with the posterior score (4), and GPT-4 is prompted to select concepts that tend to succeed the target instead. On average, for each disease, we obtain 18.5 prior and 45.0 posterior pairs of relations, which are used as labels for the performance evaluation in AUROC.

3. Results

To highlight the significance of DOME, we first numerically evaluate the embedding vectors in two EHR tasks, disease risk prediction and drug-disease relationship inference. Then, we derive and evaluate the directional KGs for the six representative phenotypes.

Table 1
Results of disease risk prediction are reported on five representative phenotypes, heart failure with reduced EF (HFrEF), chronic kidney disease (CKD), lung cancer (LC), type-2 diabetes (T2D), and Dementia (DEM) on which we evaluate the cross-site portability from MGB to UPMC. We report the AUROC in the unsupervised (supervised) scenarios.

Setting	Method	HFrEF (MGB)	CKD (MGB)	LC (MGB)	T2D (MGB)	DEM (UPMC)
6M	CODER [31]	0.589(0.830 ± 0.036)	0.563(0.739 ± 0.037)	0.793(0.865 ± 0.029)	0.692(0.852 ± 0.023)	0.578(0.822)
	GloVe [35]	0.811(0.865 ± 0.028)	0.578(0.750 ± 0.025)	0.556(0.836 ± 0.031)	0.625(0.805 ± 0.019)	0.621(0.849)
	SapBERT [32]	0.829(0.880 ± 0.032)	0.626(0.801 ± 0.033)	0.804(0.872 ± 0.040)	0.614(0.826 ± 0.023)	0.564(0.825)
	PubMedBERT [28]	0.798(0.876 ± 0.032)	0.603(0.782 ± 0.033)	0.823(0.886 ± 0.036)	0.567(0.829 ± 0.033)	0.597(0.839)
	BEHRT [16]	–(0.903 ± 0.042)	–(0.796 ± 0.042)	–(0.907 ± 0.039)	–(0.870 ± 0.038)	–(0.815)
	SGNS [47]	0.789(0.852 ± 0.023)	0.672(0.778 ± 0.028)	0.767(0.868 ± 0.032)	0.668(0.833 ± 0.015)	0.635(0.847)
	DOME	**0.854(0.910** ± 0.024)	**0.680(0.803** ± 0.021)	**0.859(0.915** ± 0.019)	**0.753(0.877** ± 0.017)	**0.687(0.866)**
12M	CODER [31]	0.592(0.825 ± 0.031)	0.561(0.758 ± 0.029)	0.782(0.873 ± 0.026)	0.674(0.860 ± 0.019)	0.596(0.825)
	GloVe [35]	0.802(0.860 ± 0.024)	0.565(0.753 ± 0.022)	0.560(0.844 ± 0.028)	0.603(0.813 ± 0.017)	0.603(0.857)
	SapBERT [32]	**0.824**(0.892 ± 0.024)	0.578(0.750 ± 0.025)	0.803(0.876 ± 0.034)	0.610(0.831 ± 0.022)	0.567(0.836)
	PubMedBERT [28]	0.795(0.869 ± 0.036)	0.609(0.776 ± 0.028)	0.818(0.892 ± 0.033)	0.567(0.827 ± 0.023)	0.576(0.823)
	BEHRT [16]	–(0.897 ± 0.046)	–(0.780 ± 0.044)	–(0.901 ± 0.045)	–(0.858 ± 0.041)	–(0.807)
	SGNS [47]	0.745(0.834 ± 0.021)	0.669(0.781 ± 0.027)	0.771(0.871 ± 0.029)	0.672(0.836 ± 0.018)	0.617(0.851)
	DOME	0.806(**0.901** ± 0.019)	**0.681(0.800** ± 0.023)	**0.863(0.923** ± 0.025)	**0.755(0.880** ± 0.015)	**0.674(0.858)**

3.1. Disease risk prediction

DOME consistently attains significant improvements over the compared embedding methods in both supervised and unsupervised scenarios, as summarized in Table 1. In unsupervised disease risk prediction for the next 6 months, DOME improves upon the state-of-the-art (SOTA) embedding vectors (underlined in Table 1) with relative gains of 3.0%, 1.2%, 4.4%, and 8.8% in HFrEF, CKD, LC, and T2D, respectively. In the supervised counterpart, the relative increments are 3.4%, 0.2%, 0.8%, and 0.8% correspondingly. In terms of unsupervised disease risk prediction in the next 12 months, DOME also outperforms most of the competing methods. DOME boosts the prediction performance relatively by 1.8%, 5.5%, and 12.0% for CKD, LC, and T2D, respectively. Relative improvements of 2.4%, 2.4%, and 2.3% are also correspondingly observed in the supervised counterpart for CKD, LC, and T2D. On the risk prediction of DEM, DOME achieves much stronger cross-site portability, outperforming the second-best model by 8.2% and 2.0% when predicting the next 6 months.

3.2. Drug-disease relationship inference

3.2.1. Drug side effects prediction

We investigate DOME's ability to discern temporally directional drug-disease relationships, with a focus on drug side effect prediction and drug indication prediction. As presented in Table 2, DOME exhibits superior performance compared to other competing embeddings in the prediction of drug side effects, especially in the unsupervised scenario. DOME improves the AUROC by 10.8% relative to the second-best model, GloVe, and increases recall@200 by 16.3% in the unsupervised scenario, while also yielding a gain of 3.2% in AUROC in the supervised setting. Notably, DOME demonstrates enlarged advantages over other methods when the study is limited to cancer drugs. In the unsupervised setting, DOME exhibits an elevated gain of 12.1% in AUROC and a 20.0% increase in recall@200. These results are anticipated because cancer drugs are more likely to trigger side effects, enabling DOME to effectively capture these intricate relationships. On the other hand, competing models such as CODER and SGNS display significantly lower performance in predicting drug side effects, with performance close to random guessing. This discrepancy suggests that these models struggle to encapsulate the temporal dependencies between drugs and their respective side effects.

3.2.2. Drug indications prediction

The results of the drug indication prediction are summarized in Table 3. DOME outperforms most of the compared embedding vectors. On the whole dataset, DOME improves the SOTA method by 6.6% in AUROC and by 4.8% in recall@30 in the unsupervised scenario, and by 1.4% in AUROC in the supervised counterpart. The performance of GloVe is comparable to SGNS in AUROC, but its recall@30 is significantly lower. When the analysis is limited to cancer drugs, all methods exhibit performance improvements. Even though the improvement in DOME is lessened, the increase in AUROC remains as high as 1.5% and 1.3% in the unsupervised and supervised settings, respectively.

3.3. Directional knowledge graph

In this section, we illustrate the directional KG derived from the directional embedding vectors. The directional edges indicate temporally directional relationships between the medical concepts. As specified in Section 2.3, we evaluate the directional relationship extraction on six aforementioned phenotypes, T2D, LC, PD, RA, CKD, and HFrEF.

3.3.1. Evaluation of directional relationship

DOME consistently achieves the highest AUROC in the prediction of both priors and posteriors for all the six phenotypes, when compared with SGNS, GloVe, and three more recent models based on BERT, including PubMedBERT [28], SapBERT [32] and CODER [31] (Table 4). This indicates the effectiveness of DOME in learning the directional relationships between concepts for different diseases. DOME outperforms non-directional methods since it encodes temporal information and exploits the assumption that the risk factors tend to occur before the disease and the treatment or complications to succeed the disease.

3.3.2. Visualization of knowledge graph

The visualizations of the directional KG with HFrEF, CKD, and hypertensive heart and/or renal disease as centered concepts are presented in Fig. 3, and KGs of the other four phenotypes are provided in Supplementary Note 7. For clarity, only the top 15 predictions are displayed and PheCode is exclusively shown in priors.

The inferred directional relationships are predominantly supported by relevant literature. Specifically, for HFrEF, most of the inferred priors are cardiovascular diseases that tend to progress into HF, including primary/intrinsic cardiomyopathies [61], nonrheumatic tricuspid valve disorders [62], and premature beats [63]. The predicted posteriors entail medications, complications, and lab tests. In particular, the top predicted posteriors for HFrEF include torsemide, milrinone, and bumetanide, all of which are indicated for HF [64]. The top predicted complications contain cardiogenic shock, which is one of the most severe manifestations of HF [65], and myocarditis [66]. The top lab tests contain heparin and hemoglobin. Heparin is recommended in the HF treatment guidelines [67]. The hemoglobin assessment is critical given that HF patients may exhibit anomalies in oxygen-hemoglobin binding to ensure sufficient oxygen delivery to the myocardial tissue [68].

For CKD, most of the inferred priors are risk factors such as other hypertensive complications [69], abnormal results of function study of kidney [60], as well as proteinuria [70]. The top inferred posteriors mainly consist of treatments and complications of CKD. The

Table 2
Embedding vectors for drug side effects prediction based on annotations from the SIDER 4.1 dataset [53] (U: unsupervised; S: supervised).

Method	Overall			Cancer drugs		
	AUROC(U)	recall@200(U)	AUROC(S)	AUROC(U)	recall@200(U)	AUROC(S)
CODER [31]	0.493	0.029	0.835 ± 0.039	0.474	0.019	0.823 ± 0.046
SGNS [47]	0.547	0.094	0.853 ± 0.038	0.555	0.105	0.860 ± 0.043
SapBERT [32]	0.548	0.176	0.869 ± 0.052	0.529	0.157	0.828 ± 0.037
PubMedBERT [28]	0.570	0.209	0.883 ± 0.044	0.565	0.230	0.834 ± 0.041
GloVe [35]	0.576	0.110	0.861 ± 0.029	0.601	0.115	0.873 ± 0.048
DOME	**0.638**	**0.243**	**0.911** ± 0.027	**0.674**	**0.276**	**0.919** ± 0.036

Table 3
Embedding vectors for drug indication prediction based on annotations from PrimeKG [55] (U: unsupervised; S: supervised).

Method	Overall			Cancer drugs		
	AUROC(U)	recall@30(U)	AUROC(S)	AUROC(U)	recall@30(U)	AUROC(S)
CODER [31]	0.516	0.002	0.893 ± 0.031	0.578	0.0	0.873 ± 0.042
SapBERT [32]	0.733	0.278	0.905 ± 0.041	0.784	0.233	0.869 ± 0.051
PubMedBERT [28]	0.745	0.320	0.922 ± 0.028	0.856	0.445	0.879 ± 0.038
SGNS [47]	0.822	0.372	0.928 ± 0.024	0.922	**0.517**	0.943 ± 0.031
GloVe [35]	0.832	0.226	0.875 ± 0.021	0.885	0.353	0.881 ± 0.029
DOME	**0.887**	**0.390**	**0.941** ± 0.020	**0.936**	0.513	**0.955** ± 0.026

Table 4
Numeric evaluation of embedding vectors on distinguishing the prior from posterior directional relationships involving type-2 diabetes (T2D), lung cancer (LC), Parkinson's disease (PD), rheumatoid arthritis (RA), chronic kidney disease (CKD), and heart failure with reduced EF (HFrEF).

Method	Prior						Posterior					
	T2D	LC	PD	RA	CKD	HFrEF	T2D	LC	PD	RA	CKD	HFrEF
SGNS [47]	0.821	0.833	0.699	0.836	0.930	0.961	0.888	0.964	0.944	0.934	0.876	0.964
GloVe [35]	0.658	0.859	0.803	0.853	0.774	0.940	0.655	0.752	0.876	0.832	0.787	0.925
SapBERT [32]	0.829	0.862	0.696	0.780	0.794	0.894	0.821	0.729	0.783	0.788	0.820	0.752
CODER [31]	0.861	0.820	0.905	0.871	0.910	0.912	0.713	0.868	0.927	0.797	0.890	0.805
PubMedBERT [28]	0.766	0.898	0.791	0.750	0.864	0.888	0.772	0.905	0.840	0.926	0.824	0.734
DOME	**0.898**	**0.972**	**0.942**	**0.895**	**0.940**	**0.969**	**0.905**	**0.992**	**0.979**	**0.977**	**0.908**	**0.970**

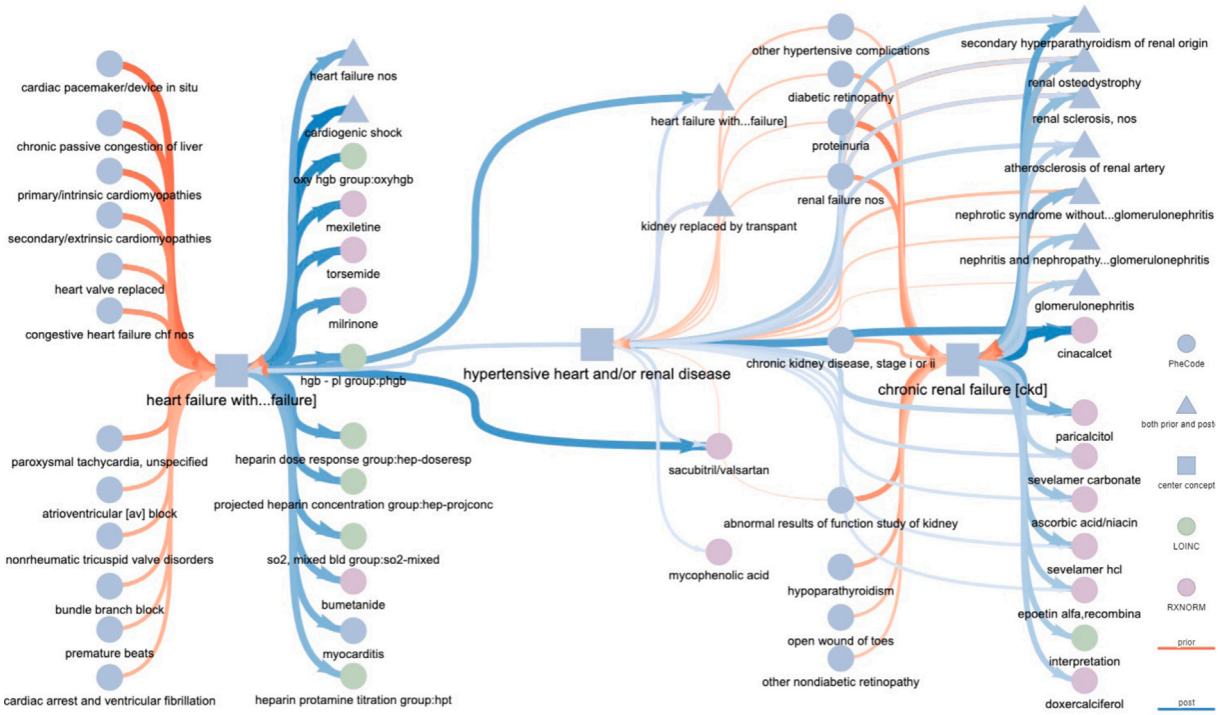

Fig. 3. Visualization of the derived directional knowledge graphs with HFrEF and CKD as the center concepts (in square shape). The red and blue edges correspondingly denote prior and posterior relationships with respect to the center concepts. The edge width (wider for a higher score) and color intensity (darker for a higher score) are proportional to the derived prior or posterior scores of the concept to the target. If a context concept appears both among the top priors and posteriors, it is shaped in a triangle.

former encompasses the medications for CKD or its common complications, such as cinacalcet, paricalcitol, and sevelamer hcl [64]. The latter includes atherosclerosis of the renal artery, renal sclerosis [71], nephrotic syndrome, nephritis, and glomerulonephritis [60]. Interestingly, these complications are also among the top inferred priors. Damages to glomerular, tubular, or renovascular components typically lead to CKD. However, regardless of its origin, CKD consequently impacts the glomerular, tubular, vascular, or interstitial compartments of the kidneys, progressing to a similar phenotype in spite of the underlying etiology. For instance, glomerular diseases culminate in tubulointerstitial lesions, and inversely, diseases affecting tubulointerstitial ultimately result in glomerular damage [72].

Moreover, DOME reveals that hypertensive heart and/or renal disease functions as the intermediate medical concept linking CKD to HFrEF. Hypertension can act as both a cause and a consequence of impaired kidney function, while elevated blood pressure remains a leading cause of HF [73]. Several of the posteriors of CKD constitute priors of hypertensive heart and/or renal disease, such as atherosclerosis of renal artery [71,74], nephrotic syndrome, nephritis, and glomerulonephritis [72,75]. Hypertensive heart and/or renal diseases share several risk factors with CKD, including other hypertensive complications [69,76], proteinuria [77,78], and abnormal results of function study of kidney. Complications include secondary hyperparathyroidism of renal origin [79,80], renal sclerosis, atherosclerosis of renal artery [71,81] along with certain medications, including cocacalet, paricalcitol and sevelamer carbonate [64], also present shared posteriors.

4. Discussions

In our study, we introduce a new approach, named DOME, for learning directional medical embedding vectors from summary-level EHR data that capture the temporal dependencies between EHR concepts. Unlike existing embedding vectors that only identify semantic similarities between EHR concepts, DOME leverages an asymmetric summary-level co-occurrence matrix to preserve the directional connections among these concepts. To effectively exploit this asymmetric structure, DOME performs joint matrix factorizations, enabling the simultaneous extraction of both semantic and directional contextual embeddings.

DOME offers inherent advantages over existing embedding techniques based on deep-learning models, such as the BEHRT [16], CLMBR [29] and Med-BERT [17], which rely on patient-level data and consequently raise privacy concerns [82]. Due to privacy issues, these pre-trained models are typically not allowed for public release as community resources. In contrast, DOME creates medical embedding vectors using summary-level EHR data, which protects individual patient privacy and enables the embedding vectors to be shared publicly. Moreover, DOME supports collaborative learning across multiple institutions without risking patient privacy by utilizing a co-training approach on the summary-level co-occurrence matrix [23,83]. For future work, it is promising to develop deep models that get rid of patient-level data and are trained also using summary-level data.

Uncovering the directional relationship among healthcare concepts without supervision is crucial for multiple healthcare studies. DOME exploits the temporal connections between EHR concepts to infer disease risk factors and comorbidities, revealing how diseases evolve over time. It approaches risk prediction as an inference of conditional relationships, enabling the modeling of disease risks in an unsupervised manner. In comparison to the deep-learning-based models [16–18] which typically require large-scale expert annotations for supervised training, DOME is clinically significant as such annotations are costly and not widely accessible especially when involving novel diseases such as COVID-19.

There are two potential avenues to explore to further improve DOME. One is to use time-adaptive windows. Currently, the time scope of the learned temporal dependency is determined by the pre-selected time windows of the co-occurrence matrix. However, to model the clinical effects temporally for different diseases or drugs, different time scales may be required. For chronic diseases, month-level time windows may suffice, whereas, for acute side effects, day-level time windows would be required to capture the temporal dependency between healthcare concepts. Therefore, expert knowledge of the specific downstream task is needed to select an appropriate time window. The other is to integrate granular time information into the embedding vectors. For example, beyond simply knowing the comorbidities of a disease, it would be beneficial to know the precise time gap within which the comorbidities are likely to occur. Such timing information is crucial for clinical decisions and treatments.

CRediT authorship contribution statement

Jun Wen: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Hao Xue:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology. **Everett Rush:** Data curation. **Vidul A. Panickan:** Data curation. **Tianrun Cai:** Data curation. **Doudou Zhou:** Data curation. **Yuk-Lam Ho:** Data curation. **Lauren Costa:** Project administration, Data curation. **Edmon Begoli:** Project administration, Data curation. **Chuan Hong:** Data curation. **J. Michael Gaziano:** Resources, Project administration, Funding acquisition. **Kelly Cho:** Resources, Project administration, Funding acquisition. **Katherine P. Liao:** Supervision, Methodology, Funding acquisition, Conceptualization. **Junwei Lu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Tianxi Cai:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jbi.2024.104768.

References

[1] Abel N Kho, Jennifer A Pacheco, Peggy L Peissig, Luke Rasmussen, Katherine M Newton, Noah Weston, Paul K Crane, Jyotishman Pathak, Christopher G Chute, Suzette J Bielinski, et al., Electronic medical records for genetic research: results of the eMERGE consortium, Sci. Transl. Med. 3 (79) (2011) 79re1.

[2] Isaac S. Kohane, Using electronic health records to drive discovery in disease genomics, Nature Rev. Genet. 12 (6) (2011) 417–428.

[3] RA Wilke, H Xu, JC Denny, DM Roden, RM Krauss, CA McCarty, RL Davis, T Skaar, J Lamba, G Savova, The emerging role of electronic medical records in pharmacogenomics, Clin. Pharmacol. Ther. 89 (3) (2011) 379–386.

[4] Sheng Yu, Tianrun Cai, Tianxi Cai, NILE: fast natural language processing for electronic health records, 2013, arXiv preprint arXiv:1311.6063.

[5] Olivier Bodenreider, The unified medical language system (UMLS): integrating biomedical terminology, Nucleic Acids Res. 1;32 (2004) D267–D270.

[6] Young Juhn, Hongfang Liu, Artificial intelligence approaches using natural language processing to advance EHR-based clinical research, J. Allergy Clin. Immunol. 145 (2) (2020) 463–469.

[7] Jue Hou, Rachel Zhao, Jessica Gronsbell, Brett K Beaulieu-Jones, Griffin Webber, Thomas Jemielita, Shuyan Wan, Chuan Hong, Yucong Lin, Tianrun Cai, Jun Wen, Vidul Panickan, Clara-Lea Bonzel Bonzel, Kai-Li Liaw, Katherine Liao, Tianxi Cai, Generate analysis-ready data for real-world evidence: tutorial for harnessing electronic health records with advanced informatic technologies, J. Med. Internet Res. 25 (2023) e45662.

[8] Jun Wen, Jue Hou, Clara-Lea Bonzel, Yihan Zhao, Victor M Castro, Vivian S Gainer, Dana Weisenfeld, Tianrun Cai, Yuk-Lam Ho, Vidul A Panickan, et al., LATTE: Label-efficient incident phenotyping from longitudinal electronic health records, Patterns 5 (1) (2024).

[9] Wei-Qi Wei, Joshua C. Denny, Extracting research-quality phenotypes from electronic health records to support precision medicine, Genome Med. 7 (1) (2015) 1–14.

[10] Po-Yen Wu, Chih-Wen Cheng, Chanchala D Kaddi, Janani Venugopalan, Ryan Hoffman, May D Wang, –Omic and electronic health record big data analytics for precision medicine, IEEE Trans. Biomed. Eng. 64 (2) (2016) 263–273.

[11] Patrick J O'Connor, JoAnn M Sperl-Hillen, William A Rush, Paul E Johnson, Gerald H Amundson, Stephen E Asche, Heidi L Ekstrom, Todd P Gilmer, Impact of electronic health record clinical decision support on diabetes care: a randomized trial, Ann. Fam. Med. 9 (1) (2011) 12–21.

[12] Rachel Gold, Christina Sheppler, Danielle Hessler, Arwen Bunce, Erika Cottrell, Nadia Yosuf, Maura Pisciotta, Rose Gunn, Michael Leo, Laura Gottlieb, et al., Using electronic health record–based clinical decision support to provide social risk–Informed care in community health centers: Protocol for the design and assessment of a clinical decision support tool, JMIR Res. Protoc. 10 (10) (2021) e31733.

[13] Youngduck Choi, Chill Yi-I Chiu, David Sontag, Learning low-dimensional representations of medical concepts, AMIA Summits Transl. Sci. Proc. 2016 (2016) 41.

[14] Li Wang, Qinghua Wang, Heming Bai, Cong Liu, Wei Liu, Yuanpeng Zhang, Lei Jiang, Huji Xu, Kai Wang, Yunyun Zhou, EHR2Vec: representation learning of medical concepts from temporal patterns of clinical notes based on self-attention mechanism, Front. Genet. 11 (2020) 630.

[15] Wael Farhan, Zhimu Wang, Yingxiang Huang, Shuang Wang, Fei Wang, Xiaoqian Jiang, et al., A predictive model for medical events based on contextual embedding of temporal sequences, JMIR Med. Inform. 4 (4) (2016) e5977.

[16] Yikuan Li, Shishir Rao, José Roberto Ayala Solares, Abdelaali Hassaine, Rema Ramakrishnan, Dexter Canoy, Yajie Zhu, Kazem Rahimi, Gholamreza Salimi-Khorshidi, BEHRT: transformer for electronic health records, Sci. Rep. 10 (1) (2020) 1–12.

[17] Laila Rasmy, Yang Xiang, Ziqian Xie, Cui Tao, Degui Zhi, Med-BERT: pretrained contextualized embeddings on large-scale structured electronic health records for disease prediction, NPJ Digit. Med. 4 (1) (2021) 1–13.

[18] Kexin Huang, Jaan Altosaar, Rajesh Ranganath, ClinicalBERT: Modeling clinical notes and predicting hospital readmission, 2019, arXiv preprint arXiv:1904.05342.

[19] Edward Choi, Cao Xiao, Walter Stewart, Jimeng Sun, MiME: Multilevel medical embedding of electronic health records for predictive healthcare, Adv. Neural Inf. Process. Syst. 31 (2018).

[20] Emily Getzen, Yucheng Ruan, Lyle Ungar, Qi Long, Mining for health: A comparison of word embedding methods for analysis of EHRs data, medRxiv (2022).

[21] Charlotte A Nelson, Riley Bove, Atul J Butte, Sergio E Baranzini, Embedding electronic health records onto a knowledge network recognizes prodromal features of multiple sclerosis and predicts diagnosis, J. Am. Med. Inform. Assoc. 29 (3) (2022) 424–434.

[22] Junyu Luo, Muchao Ye, Cao Xiao, Fenglong Ma, HiTANet: Hierarchical time-aware attention networks for risk prediction on electronic health records, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2020, pp. 647–656.

[23] Chuan Hong, Everett Rush, Molei Liu, Doudou Zhou, Jiehuan Sun, Aaron Sonabend, Victor M Castro, Petra Schubert, Vidul A Panickan, Tianrun Cai, et al., Clinical knowledge extraction via sparse embedding regression (KESER) with multi-center large scale electronic health record data, medRxiv (2021).

[24] Jun Wen, Xiang Zhang, Everett Rush, Vidul A Panickan, Xingyu Li, Tianrun Cai, Doudou Zhou, Yuk-Lam Ho, Lauren Costa, Edmon Begoli, et al., Multimodal representation learning for predicting molecule–disease relations, Bioinformatics 39 (2) (2023) btad085.

[25] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, Illia Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).

[26] Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, BERT: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers), 2019, pp. 4171–4186.

[27] Iz Beltagy, Kyle Lo, Arman Cohan, SciBERT: A pretrained language model for scientific text, in: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP), 2019, pp. 3615–3620.

[28] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, Hoifung Poon, Domain-specific language model pretraining for biomedical natural language processing, ACM Trans. Comput. Healthc. (HEALTH) 3 (1) (2021) 1–23.

[29] Lin Lawrence Guo, Ethan Steinberg, Scott Lanyon Fleming, Jose Posada, Joshua Lemmon, Stephen R Pfohl, Nigam Shah, Jason Fries, Lillian Sung, EHR foundation models improve robustness in the presence of temporal distribution shift, Sci. Rep. 13 (1) (2023) 3767.

[30] Zhichao Yang, Avijit Mitra, Weisong Liu, Dan Berlowitz, Hong Yu, TransformEHR: transformer-based encoder-decoder generative model to enhance prediction of disease outcomes using electronic health records, Nature Commun. 14 (1) (2023) 7857.

[31] Zheng Yuan, Zhengyun Zhao, Haixia Sun, Jiao Li, Fei Wang, Sheng Yu, CODER: Knowledge-infused cross-lingual medical term embedding for term normalization, J. Biomed. Inform. 126 (2022) 103983.

[32] Fangyu Liu, Ehsan Shareghi, Zaiqiao Meng, Marco Basaldella, Nigel Collier, Self-alignment pretraining for biomedical entity representations, in: Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, 2021, pp. 4228–4238.

[33] Tong Wu, Yunlong Wang, Yue Wang, Emily Zhao, Yilian Yuan, Leveraging graph-based hierarchical medical entity embedding for healthcare applications, Sci. Rep. 11 (1) (2021) 5858.

[34] Martin Müller, Marcel Salathé, Per E. Kummervold, COVID-Twitter-BERT: A natural language processing model to analyse covid-19 content on twitter, Front. Artif. Intell. 6 (2023) 1023281.

[35] Jeffrey Pennington, Richard Socher, Christopher D. Manning, Glove: Global vectors for word representation, in: Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing, EMNLP, 2014, pp. 1532–1543.

[36] Doudou Zhou, Ziming Gan, Xu Shi, Alina Patwari, Everett Rush, Clara-Lea Bonzel, Vidul A Panickan, Chuan Hong, Yuk-Lam Ho, Tianrun Cai, Lauren Costa, Xiaoou Li, Victor M Castro, Shawn N Murphy, Gabriel Brat, Griffin Weber, Paul Avillach, J Michael Gaziano, Kelly Cho, Katherine P Liao, Junwei Lu, Tianxi Cai, Multiview incomplete knowledge graph integration with application to cross-institutional EHR data harmonization, J. Biomed. Inform. 133 (2022) 104147.

[37] Henry Völzke, Multicausality in fatty liver disease: is there a rationale to distinguish between alcoholic and non-alcoholic origin? World J. Gastroenterol.: WJG 18 (27) (2012) 3492.

[38] Alberto Francisco Rubio-Guerra, Leticia Rodriguez-Lopez, German Vargas-Ayala, Saul Huerta-Ramirez, David Castro Serna, Jose Juan Lozano-Nuevo, Depression increases the risk for uncontrolled hypertension, Exp. Clin. Cardiol. 18 (1) (2013) 10.

[39] Gregory J Moore, Frank P MacMaster, Carol Stewart, David R Rosenberg, Case study: caudate glutamatergic changes with paroxetine therapy for pediatric obsessive-compulsive disorder, J. Am. Acad. Child Adolesc. Psychiatry 37 (6) (1998) 663–667.

[40] Daniel A Geller, Joseph Biederman, S Evelyn Stewart, Benjamin Mullin, Colleen Farrell, Karen Dineen Wagner, Graham Emslie, David Carpenter, Impact of comorbidity on treatment response to paroxetine in pediatric obsessive-compulsive disorder: Is the use of exclusion criteria empirically supported in randomized clinical trials? J. Child Adolesc. Psychopharmacol. 13 (2, Supplement 1) (2003) 19–29.

[41] Alexandra L Young, Razvan V Marinescu, Neil P Oxtoby, Martina Bocchetta, Keir Yong, Nicholas C Firth, David M Cash, David L Thomas, Katrina M Dick, Jorge Cardoso, et al., Uncovering the heterogeneity and temporal complexity of neurodegenerative diseases with Subtype and Stage Inference, Nature Commun. 9 (1) (2018) 4273.

[42] Rachel Lowe, Trevor C Bailey, David B Stephenson, Richard J Graham, Caio AS Coelho, Marilia Sa Carvalho, Christovam Barcellos, Spatio-temporal modelling of climate-sensitive disease risk: Towards an early warning system for dengue in Brazil, Comput. Geosci. 37 (3) (2011) 371–381.

[43] Xing Song, Lemuel R Waitman, SL Alan, David C Robbins, Yong Hu, Mei Liu, et al., Longitudinal risk prediction of chronic kidney disease in diabetic patients using a temporal-enhanced gradient boosting machine: retrospective cohort study, JMIR Med. Inform. 8 (1) (2020) e15510.

[44] Anders Boeck Jensen, Pope L Moseley, Tudor I Oprea, Sabrina Gade Ellesøe, Robert Eriksson, Henriette Schmock, Peter Bjødstrup Jensen, Lars Juhl Jensen, Søren Brunak, Temporal disease trajectories condensed from population-wide registry data covering 6.2 million patients, Nature Commun. 5 (1) (2014) 4022.

[45] Hossein Estiri, Zachary H Strasser, Jeffy G Klann, Pourandokht Naseri, Kavishwar B Wagholikar, Shawn N Murphy, Predicting COVID-19 mortality with electronic medical records, NPJ Digit. Med. 4 (1) (2021) 1–10.

[46] Yan Song, Shuming Shi, Jing Li, Haisong Zhang, Directional skip-gram: Explicitly distinguishing left and right context for word embeddings, in: Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers), 2018, pp. 175–180.

[47] Omer Levy, Yoav Goldberg, Neural word embedding as implicit matrix factorization, Adv. Neural Inf. Process. Syst. 27 (2014).

[48] Tomas Mikolov, Ilya Sutskever, Kai Chen, Greg S Corrado, Jeff Dean, Distributed representations of words and phrases and their compositionality, Adv. Neural Inf. Process. Syst. 26 (2013).

[49] Ilya Sutskever, James Martens, George Dahl, Geoffrey Hinton, On the importance of initialization and momentum in deep learning, in: International Conference on Machine Learning, PMLR, 2013, pp. 1139–1147.

[50] Benjamin A Goldstein, Ann Marie Navar, Michael J Pencina, John Ioannidis, Opportunities and challenges in developing risk prediction models with electronic health records data: A systematic review, J. Am. Med. Inform. Assoc. 24 (1) (2017) 198–208.

[51] Lorna Hazell, Saad A.W. Shakir, Under-reporting of adverse drug reactions, Drug Saf. 29 (5) (2006) 385–396.

[52] Janet Sultana, Paola Cutroneo, Gianluca Trifirò, Clinical and economic burden of adverse drug reactions, J. Pharmacol. Pharmacother. 4 (1_suppl) (2013) S73–S77.

[53] Michael Kuhn, Ivica Letunic, Lars Juhl Jensen, Peer Bork, The SIDER database of drugs and side effects, Nucleic Acids Res. 44 (D1) (2016) D1075–D1079.

[54] Y. Yu, K. Huang, C. Zhang, L.M. Glass, J. Sun, C. Xiao, SumGNN: Multi-typed drug interaction prediction via efficient knowledge graph summarization, Bioinformatics (2021).

[55] Payal Chandak, Kexin Huang, Marinka Zitnik, Building a knowledge graph to enable precision medicine, Sci. Data 10 (1) (2023) 67.

[56] Mei Liu, Yonghui Wu, Yukun Chen, Jingchun Sun, Zhongming Zhao, Xue-wen Chen, Michael Edwin Matheny, Hua Xu, Large-scale prediction of adverse drug reactions using chemical, biological, and phenotypic properties of drugs, J. Am. Med. Inform. Assoc. 19 (e1) (2012) e28–e35.

[57] Emir Muñoz, Vít Nováček, Pierre-Yves Vandenbussche, Facilitating prediction of adverse drug reactions by using knowledge graphs and multi-label learning models, Brief. Bioinform. 20 (1) (2019) 190–202.

[58] Ali Shojaie, Emily B. Fox, Granger causality: A review and recent advances, Annu. Rev. Stat. Appl. 9 (2022) 289–319.

[59] OpenAI, GPT-4 technical report, ArXiv (2023) https://openai.com/research/gpt-4.

[60] Mayo Foundation for Medical Education and Research, Mayo clinic patient care and health information, 2023, URL https://www.mayoclinic.org/patient-care-and-health-information.

[61] Petar M. Seferović, Marija Polovina, Johann Bauersachs, Michael Arad, Tuvia Ben Gal, Lars H Lund, Stephan B Felix, Eloisa Arbustini, Alida LP Caforio, Dimitrios Farmakis, et al., Heart failure in cardiomyopathies: A position paper from the Heart Failure Association of the European Society of Cardiology, Eur. J. Heart Fail. 21 (5) (2019) 553–576.

[62] Avinoam Shiran, Alex Sagie, Tricuspid regurgitation in mitral valve disease: incidence, prognostic implications, mechanism, and management, J. Am. Coll. Cardiol. 53 (5) (2009) 401–408.

[63] Worawan B Limpitikul, Thomas A Dewland, Eric Vittinghoff, Elsayed Soliman, Gregory Nah, Christina Fang, David S Siscovick, Bruce M Psaty, Nona Sotoodehnia, Susan Heckbert, et al., Premature ventricular complexes and development of heart failure in a community-based population, Heart 108 (2) (2022) 105–110.

[64] National Library of Medicine, MedlinePlus, 2023, URL https://medlineplus.gov/.

[65] Cesar Y. Guerrero-Miranda, Shelley A. Hall, Cardiogenic shock in patients with advanced chronic heart failure, Methodist DeBakey Cardiovasc. J. 16 (1) (2020) 22.

[66] P van den Hoogen, F van den Akker, JC Deddens, JPG Sluijter, Heart failure in chronic myocarditis: a role for microRNAs? Curr. Genomics 16 (2) (2015) 88–94.

[67] Yasuhiro Hamatani, Takao Kato, Takeshi Morimoto, Moritake Iguchi, Hidenori Yaku, Yasutaka Inuzuka, Takeshi Kitai, Kazuya Nagao, Yodo Tamaki, Erika Yamamoto, et al., Association of intravenous heparin administration with in-hospital clinical outcomes among hospitalized patients with acute heart failure, Int. J. Cardiol. 370 (2023) 229–235.

[68] RM Bersin, M Kwasman, D Lau, C Klinski, K Tanaka, P Khorrami, T DeMarco, Ch Wolfe, K Chatterjee, Importance of oxygen-haemoglobin binding to oxygen transport in congestive heart failure, Heart 70 (5) (1993) 443–447.

[69] Seyed Mehrdad Hamrahian, Bonita Falkner, Hypertension in chronic kidney disease, Hypertension: Basic Res. Clin. Pract. (2017) 307–325.

[70] Janet B McGill, Hermann Haller, Prabir Roy-Chaudhury, Andrea Cherrington, Takashi Wada, Christoph Wanner, Linong Ji, Peter Rossing, Making an impact on kidney disease in people with type 2 diabetes: The importance of screening for albuminuria, BMJ Open Diabetes Res. Care 10 (4) (2022) e002806.

[71] Soon Hyo Kwon, Lilach O. Lerman, Atherosclerotic renal artery stenosis: current status, Adv Chronic Kidney Dis. 22 (3) (2015) 224–231.

[72] José M López-Novoa, Ana B Rodríguez-Peña, Alberto Ortiz, Carlos Martínez-Salgado, Francisco J López Hernández, Etiopathology of chronic tubular, glomerular and renovascular nephropathies: clinical implications, J. Transl. Med. 9 (2011) 1–26.

[73] Michel Burnier, Aikaterini Damianaki, Hypertension as cardiovascular risk factor in chronic kidney disease, Circ. Res. 132 (8) (2023) 1050–1063.

[74] Rohini Manaktala, Jose D. Tafur-Soto, Christopher J. White, Renal artery stenosis in the patient with hypertension: prevalence, impact and management, Integr. Blood Press. Control (2020) 71–82.

[75] Chun-Gyoo Ihm, Hypertension in chronic glomerulonephritis, Electrolytes Blood Press.: E BP 13 (2) (2015) 41.

[76] Tomasz Stompor, Agnieszka Perkowska-Ptasińska, Hypertensive kidney disease: A true epidemic or rare disease, Pol. Arch. Intern. Med. 130 (2) (2020) 130–139.

[77] Jose Luis Gorriz, Alberto Martinez-Castelao, Proteinuria: Detection and role in native renal disease progression, Transpl. Rev. 26 (1) (2012) 3–13.

[78] George Bakris, Proteinuria: A link to understanding changes in vascular compliance? Hypertension 46 (3) (2005) 473–474.

[79] Mariadelina Simeoni, Alessandra F. Perna, Giorgio Fuiano, Secondary hyperparathyroidism and hypertension: an intriguing couple, J. Clin. Med. 9 (3) (2020) 629.

[80] Elmukhtar Habas Sr, Mohsen Eledrisi, Fahmi Khan, Abdel-Naser Y Elzouki, Fahmi Y Khan, Secondary hyperparathyroidism in chronic kidney disease: pathophysiology and management, Cureus 13 (7) (2021).

[81] Alejandro R. Chade, Amir Lerman, Lilach O. Lerman, Kidney in early atherosclerosis, Hypertension 45 (6) (2005) 1042–1049.

[82] Xudong Pan, Mi Zhang, Shouling Ji, Min Yang, Privacy risks of general-purpose language models, in: 2020 IEEE Symposium on Security and Privacy, SP, IEEE, 2020, pp. 1314–1331.

[83] Doudou Zhou, Tianxi Cai, Junwei Lu, Multi-source learning via completion of block-wise overlapping noisy matrices, J. Mach. Learn. Res. 24 (221) (2023) 1–43.